	Report	Result	Size	Time	Cycles	GPU	SM Frequency	Process	Attributes
Current	double_buffer	572 - convolution3D_kernel	(24, 24, 1)x(32, 32, 1)	62.46 us	101,073	0 - NVIDIA GeForce RTX 4060 Laptop GPU	1.62 Ghz	[117477] conv3d_double_buffer	
Baseline 1	baseline	572 - convolution3D_kernel	(24, 24, 1)x(32, 32, 1)	45.06 us	72,913	0 - NVIDIA GeForce RTX 4060 Laptop GPU	1.62 Ghz	[104238] conv3d_baseline	

Summary | Details | Source | Context | Comments | Raw | Session

Compare | Tools | View | Export

GPU Speed Of Light Throughput

GPU Throughput Chart

High-level overview of the throughput for compute and memory resources of the GPU. For each unit, the throughput reports the achieved percentage of utilization with respect to the theoretical maximum. Breakdowns show the throughput for each individual sub-metric of Compute and Memory to clearly identify the highest contributor. High-level overview of the utilization for compute and memory resources of the GPU presented as a roofline chart.

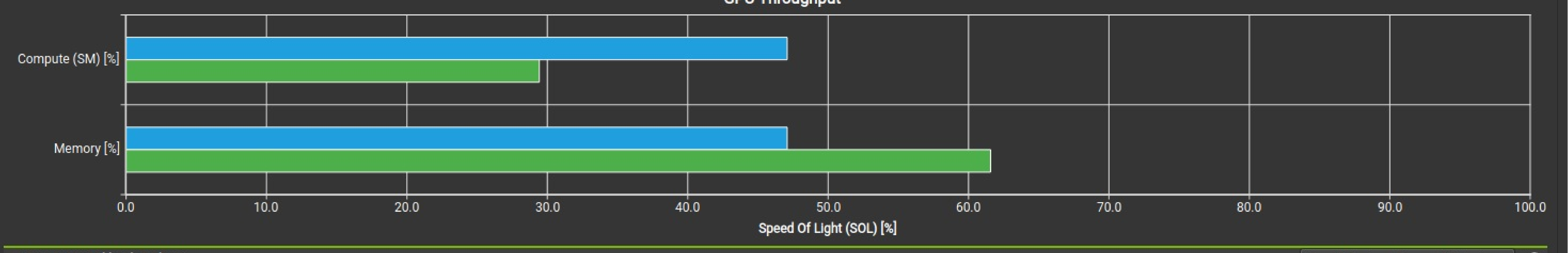
Compute (SM) Throughput [%]	47.09 (+59.99%)	Duration [us]	62.46 (+38.64%)
Memory Throughput [%]	47.09 (-23.53%)	Elapsed Cycles [cycle]	101,073 (+38.62%)
L1/TEX Cache Throughput [%]	50.55 (+55.69%)	SM Active Cycles [cycle]	94,099.46 (+42.40%)
L2 Cache Throughput [%]	13.76 (-27.55%)	SM Frequency [Ghz]	1.62 (-0.05%)
DRAM Throughput [%]	44.39 (-27.90%)	DRAM Frequency [Ghz]	7.98 (+0.08%)

Latency Issue This workload exhibits low compute throughput and memory bandwidth utilization relative to the peak performance of this device. Achieved compute throughput and/or memory bandwidth below 60.0% of peak typically indicate latency issues. Look at Scheduler Statistics and Warp State Statistics for potential reasons.

Key Performance Indicators

Roofline Analysis The ratio of peak float (FP32) to double (FP64) performance on this device is 64:1. The workload achieved 3% of this device's FP32 peak performance and 0% of its FP64 peak performance. See the Profiling Guide for more details on roofline analysis.

GPU Throughput



Compute Workload Analysis

Pipe Utilization (Elapsed Cycles)

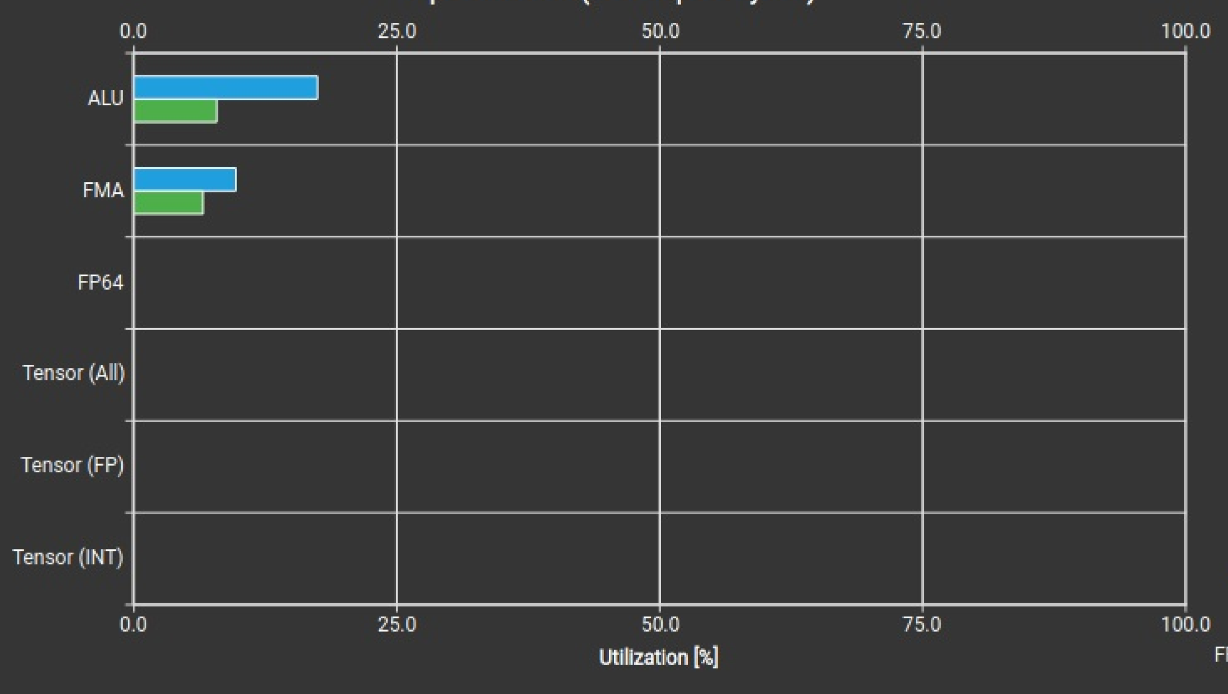
Detailed analysis of the compute resources of the streaming multiprocessors (SM), including the achieved instructions per clock (IPC) and the utilization of each available pipeline. Pipelines with very high utilization might limit the overall performance.

Executed Ipc Elapsed [inst/cycle]	1.08 (+83.68%)	SM Busy [%]	27.14 (+82.98%)
Executed Ipc Active [inst/cycle]	1.16 (+78.75%)	Issue Slots Busy [%]	27.14 (+82.98%)
Issued Ipc Active [inst/cycle]	1.17 (+78.06%)		

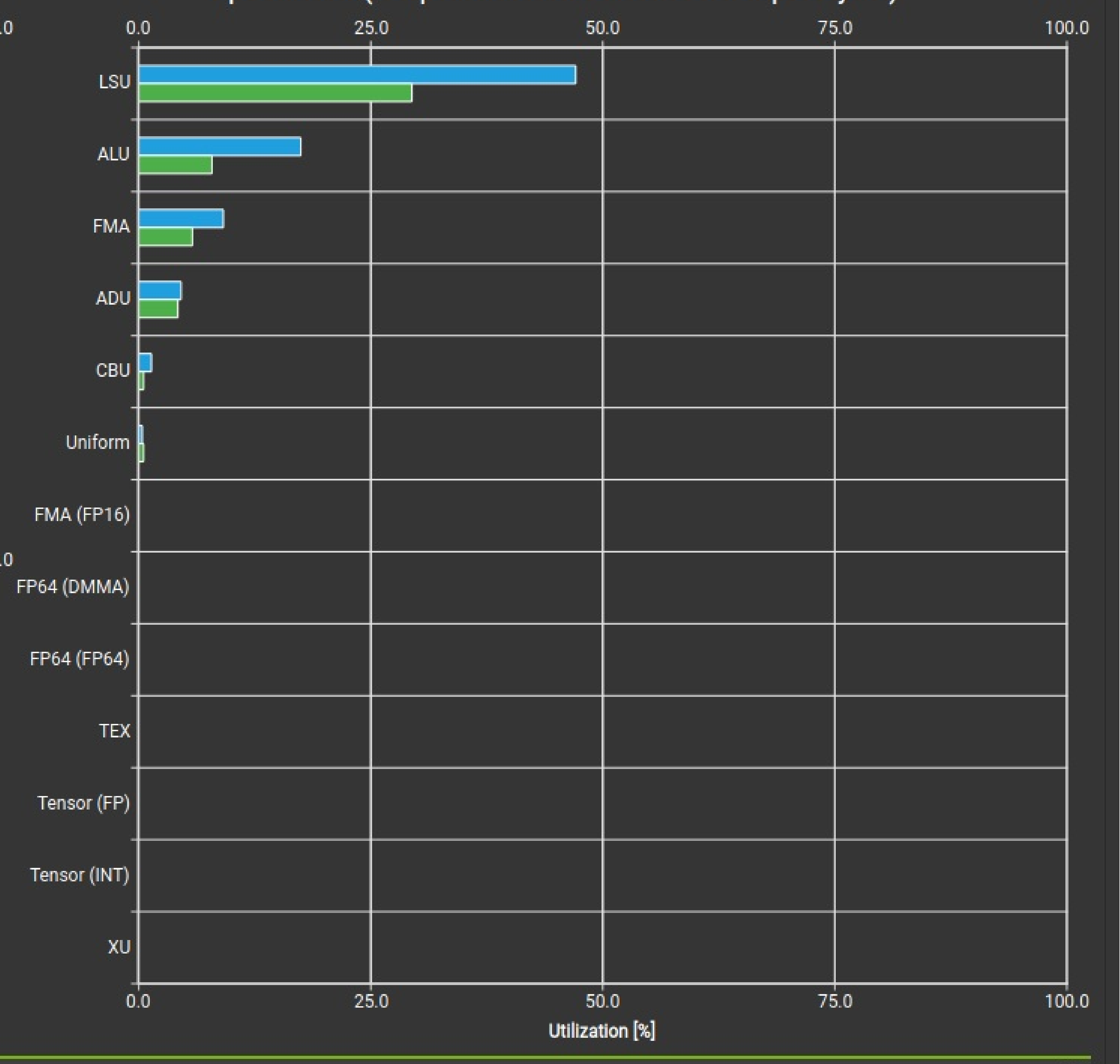
Low Utilization
Est. Local Speedup: 82.54% All compute pipelines are under-utilized. Either this workload is very small or it doesn't issue enough warps per scheduler. Check the Launch Statistics and Scheduler Statistics sections for further details.

Key Performance Indicators

Pipe Utilization (% of elapsed cycles)



Pipe Utilization (% of peak instructions executed over elapsed cycles)



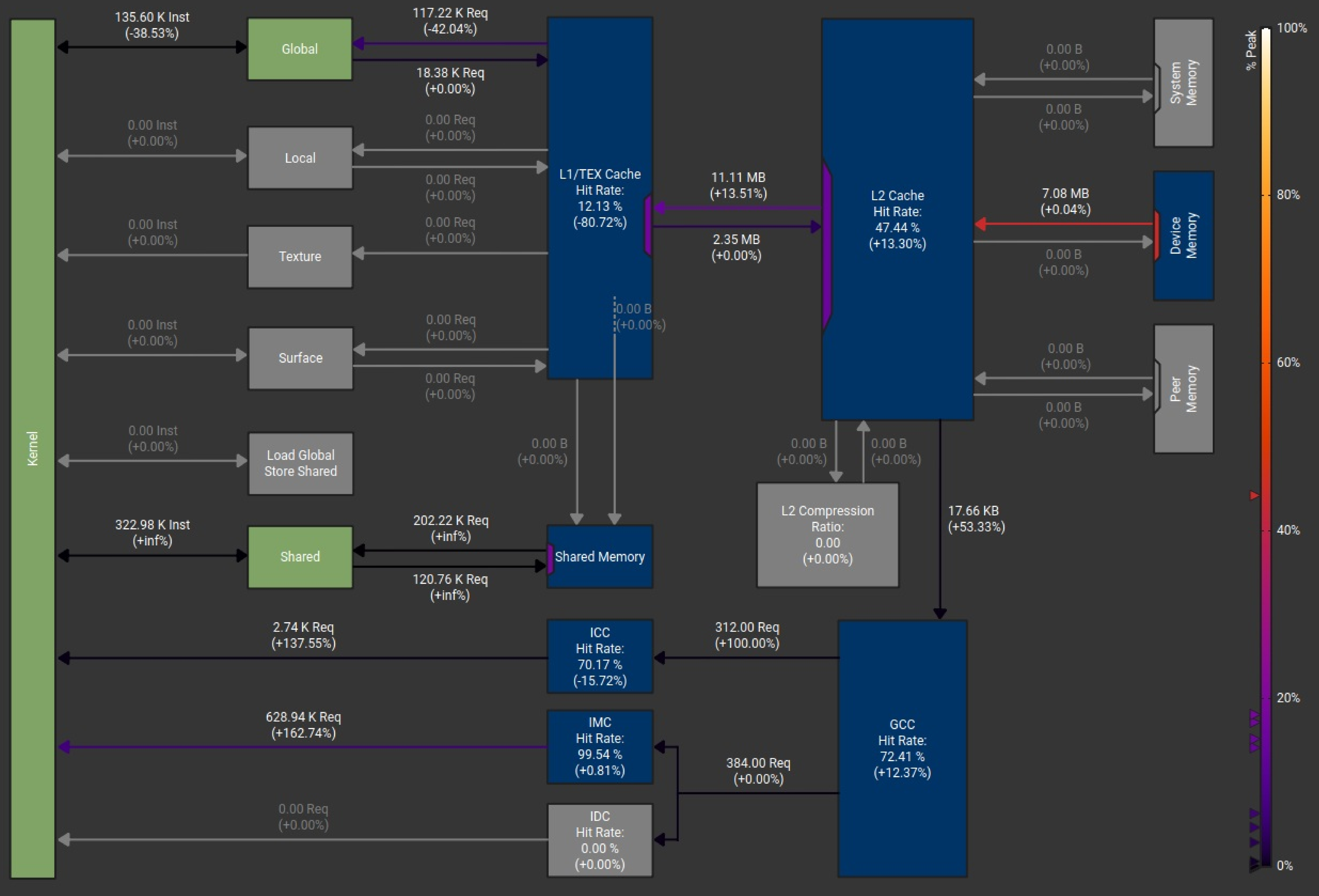
Memory Workload Analysis

Detailed analysis of the memory resources of the GPU. Memory can become a limiting factor for the overall kernel performance when fully utilizing the involved hardware units (Mem Busy), exhausting the available communication bandwidth between those units (Max Bandwidth), or by reaching the maximum throughput of issuing memory instructions (Mem Pipes Busy). Detailed chart of the memory units.

Memory Throughput [Gbyte/s]	113.42 (-27.84%)	Mem Busy [%]	26.22 (+60.12%)
L1/TEX Hit Rate [%]	12.13 (-80.72%)	Max Bandwidth [%]	47.09 (-23.53%)
L2 Hit Rate [%]	47.44 (+13.30%)	Mem Pipes Busy [%]	47.09 (+59.99%)
L2 Compression Input Sectors [sector]	0 (+0.00%)	Local Memory Spilling Requests	0 (+0.00%)
L2 Compression Ratio	0 (+0.00%)	Local Memory Spilling Request Overhead [%]	0 (+0.00%)
L2 Compression Success Rate [%]	0 (+0.00%)	L2 Persisting Size [Mbyte]	6.29 (+0.00%)

Memory Chart

Values: Transfer Size Inactivity: Greyed Out



Launch Statistics

Summary of the configuration used to launch the kernel. The launch configuration defines the size of the kernel grid, the division of the grid into blocks, and the GPU resources needed to execute the kernel. Choosing an efficient launch configuration maximizes device utilization.

Grid Size	576 (+0.00%)	Function Cache Configuration	CachePreferNone (CachePreferNone)
Registers Per Thread [register/thread]	25 (-21.88%)	Static Shared Memory Per Block [Kbyte/block]	28.56 (+inf%)
Block Size	1,024 (+0.00%)	Dynamic Shared Memory Per Block [byte/block]	0 (+0.00%)
Threads [thread]	589,824 (+0.00%)	Driver Shared Memory Per Block [Kbyte/block]	1.02 (+0.00%)
Waves Per SM	24 (+0.00%)	Shared Memory Configuration Size [Kbyte]	32.77 (+300.00%)
Uses Green Context	0 (+0.00%)	Stack Size	1,024 (+0.00%)
# SMs [SM]	24 (+0.00%)	# TPCs	12 (+0.00%)
Enabled TPC IDs	all (all)	-	-

Occupancy

% Occupancy Graphs

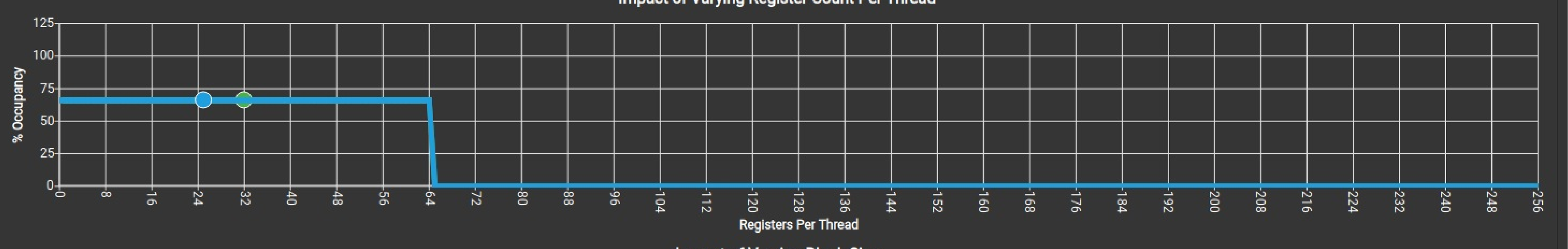
Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicates highly imbalanced workloads.

Theoretical Occupancy [%]	66.67 (+0.00%)	Block Limit Registers [block]	2 (+0.00%)
Theoretical Active Warps per SM [warp]	32 (+0.00%)	Block Limit Shared Mem [block]	1 (-87.50%)
Achieved Occupancy [%]	66.33 (+16.03%)	Block Limit Warps [block]	1 (+0.00%)
Achieved Active Warps Per SM [warp]	31.84 (+16.03%)	Block Limit SM [block]	24 (+0.00%)

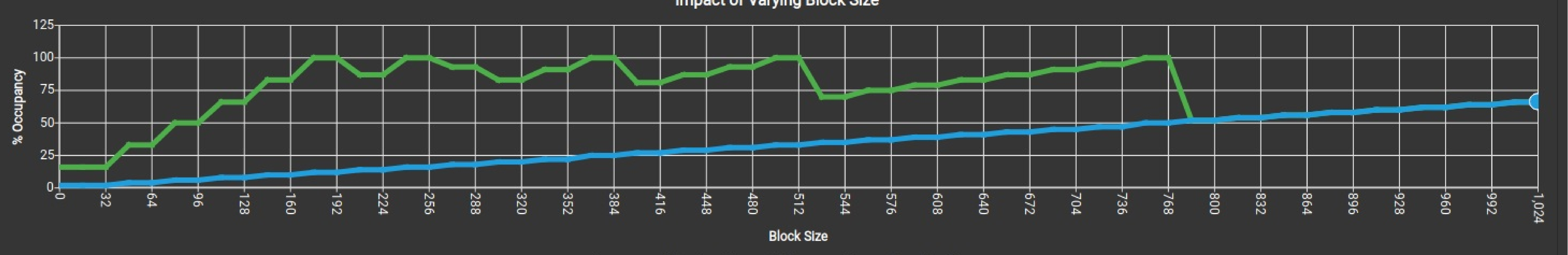
Theoretical Occupancy
Est. Local Speedup: 33.33% The 8.00 theoretical warps per scheduler this kernel can issue according to its occupancy are below the hardware maximum of 12. This kernel's theoretical occupancy (66.7%) is limited by the required amount of shared memory, and the number of warps within each block.

Key Performance Indicators

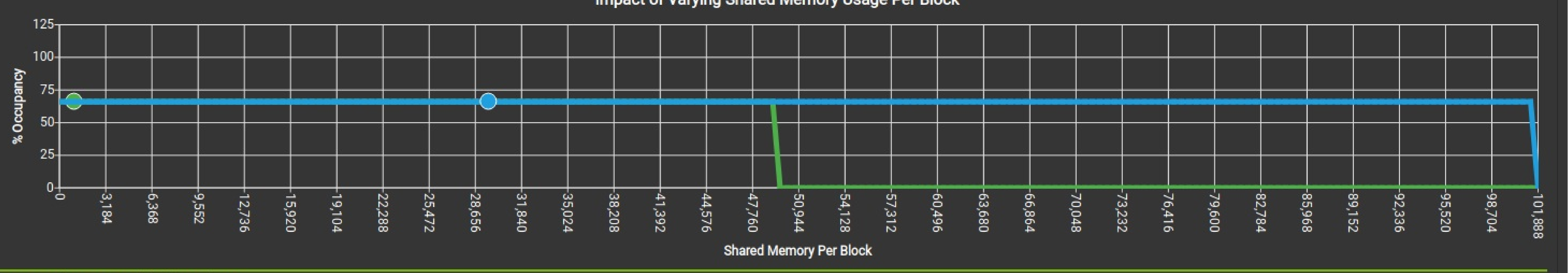
Impact of Varying Register Count Per Thread



Impact of Varying Block Size



Impact of Varying Shared Memory Usage Per Block



GPU and Memory Workload Distribution

Analysis of workload distribution in active cycles of SM, SMP, SMSP, L1 & L2 caches, and DRAM

Average SM Active Cycles [cycle]	94,099.46 (+42.40%)	Average L1 Active Cycles [cycle]	94,099.46 (+42.40%)
Average L2 Active Cycles [cycle]	73,587 (+19.80%)	Average SMSP Active Cycles [cycle]	93,376.16 (+46.86%)
Average DRAM Active Cycles [cycle]	221,388 (+0.04%)	Total SM Elapsed Cycles [cycle]	2,424,568 (+38.57%)
Total L1 Elapsed Cycles [cycle]	2,424,568 (+38.57%)	Total L2 Elapsed Cycles [cycle]	1,610,208 (+38.22%)
Total SMSP Elapsed Cycles [cycle]	9,698,272 (+38.57%)	Total DRAM Elapsed Cycles [cycle]	1,994,752 (+38.75%)

Workload Distribution

	Average	Min	Max	Sum
SM Active Cycles	94,099.46 (+42.40%)	93,648 (+43.92%)	94,550 (+41.70%)	2,258,387 (+42.40%)
SMSP Active Cycles	93,376.16 (+46.86%)	92,907 (+51.19%)	93,758 (+43.86%)	8,964,111 (+46.86%)
L1 Active Cycles	94,099.46 (+42.40%)	93,648 (+43.92%)	94,550 (+41.70%)	2,258,387 (+42.40%)
L2 Active Cycles	73,587 (+19.80%)	72,087 (+19.29%)	74,758 (+20.21%)	1,177,392 (+19.80%)
DRAM Active Cycles	221,388 (+0.04%)	221,344 (+0.06%)	221,488 (+0.07%)	885,552 (+0.04%)

Source Counters

Source metrics, including branch efficiency and sampled warp stall reasons. Warp Stall Sampling metrics are periodically sampled over the kernel runtime. They indicate when warps were stalled and couldn't be scheduled. See the documentation for a description of all stall reasons. Only focus on stalls if the schedulers fail to issue every cycle.

Branch Instructions [inst]	209,616 (+469.36%)	Branch Efficiency [%]	83 (+inf%)
Branch Instructions Ratio [%]	0.08 (+123.70%)	Avg. Divergent Branches [branches]	204 (+inf%)

Uncoalesced Global Accesses
Est. Speedup: 17.15% This kernel has uncoalesced global accesses resulting in a total of 112332 excessive sectors (23% of the total 478908 sectors). Check the L2 Theoretical Sectors Global Excessive table for the primary source locations. The CUDA Programming Guide has additional information on reducing uncoalesced device memory accesses.

Key Performance Indicators

L2 Theoretical Sectors Global Excessive

Location	Value	Value (%)
0x725aa3272ad0 in convolution3D_kernel	37,444	33
0x725aa3272ac0 in convolution3D_kernel	37,444	33
0x725aa3272ab0 in convolution3D_kernel	37,444	33
0x725aa3272e90 in convolution3D_kernel	0	0
0x725aa32727f0 in convolution3D_kernel	0	0

Warp Stall Sampling (All Samples)

Location	Value	Value (%)
0x725aa3272b30 in convolution3D_ke...	760	34
0x725aa3272c90 in convolution3D_ke...	398	18
0x725aa3272b40 in convolution3D_ke...	89	4
0x725aa3272eb0 in convolution3D_ke...	68	3
0x725aa3272b50 in convolution3D_ke...	61	3

Most Instructions Executed

Location	Value	Value (%)
0x725aa3272b80 in convolution3D_ke...	39,168	1
0x725aa3272b70 in convolution3D_ke...	39,168	1
0x725aa3272b60 in convolution3D_ke...	39,168	1
0x725aa3272b50 in convolution3D_ke...	39,168	1
0x725aa3272b40 in convolution3D_ke...	39,168	1

Still missing data? Collect the *full* metric set, enable individual sections or learn how to select individual metrics for collection.